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Future of Jobs in India: 3.0

Navigating Skills in the Digital Age

PART 1 — INTRODUCTION

Introduction to the Study

Future of Jobs in India: 3.0 — Navigating Skills in the Digital Age

MANUFACTURING

HEALTHCARE

INFRASTRUCTURE

ENERGY

FMCG

1. Introduction

The job and skilling landscape is witnessing a paradigm shift, driven by the emergence of transformative technologies and accelerating digital adoption across India and the globe. Industry 4.0 technologies — cloud computing, artificial intelligence (AI), the internet of things (IoT), robotics and more — are converging with Digital Public Infrastructure (DPI), environmental imperatives, and evolving workforce dynamics to redefine how India works, learns and hires.

This report, **Future of Jobs (FOJ) 3.0**, is the third in a series published by the Federation of Indian Chambers of Commerce & Industry (FICCI) in collaboration with Ernst & Young (EY), following earlier editions released in 2017 and 2018. It builds on that legacy with a focused, sector-level examination of five pillars of the Indian economy: **Manufacturing, Healthcare, Infrastructure, Energy and FMCG** — assessing macroeconomic trends, international benchmarks, and the diverse perspectives of employers, learners and trainers.

The study draws on both secondary research (government data, industry reports, global indices) and primary research — structured interviews and a digital survey reaching over 200 respondents, including senior industry leaders, learners, trainers, academicians and policymakers.

2. Problem Statement

India's skilling and employment ecosystem sits at an inflection point. While the country holds a significant demographic advantage — over 371 million people in the youth cohort — several structural gaps prevent this advantage from translating fully into a demographic dividend:

- ➔ **Persistent skill mismatch:** a large share of the workforce remains under-skilled relative to evolving industry requirements, and the informal sector still accounts for roughly 71.7% of all workers.
- ➔ **Low labour force participation among women:** female LFPR stands at 33.2% versus 62.2% for men, and 57% of young women are Neither in Employment, Education nor Training (NEET).
- ➔ **Weak industry-academia linkages:** vocational education is not seen as aspirational, and employers report limited awareness of government skilling schemes and misalignment between training outcomes and real industry needs.
- ➔ **Disruptive technology adoption:** AI, Generative AI and Industry 4.0 tools are transforming job roles faster than reskilling programs can respond, creating workforce resistance and a shortage of technical skills.
- ➔ **Climate and sustainability pressure:** the shift toward green jobs demands new competencies that current training infrastructure is only beginning to address.

3. Objectives of the Study

FOJ 3.0 sets out to provide a comprehensive, evidence-based view of how jobs and skills will evolve over the next two to three years, with the specific objectives of:

- ✓ Mapping macroeconomic trends, employment scenarios and international benchmarks across five key sectors.
- ✓ Assessing the impact of five converging forces — **Artificial Intelligence, Digital Public Infrastructure (DPI), Climate/Green Jobs, Workforce Mobility and Industry 4.0** — on jobs and skills.
- ✓ Capturing the perspectives of three core stakeholders: **Employers (industry), Learners** (in education, upskilling and reskilling) and **Trainers** (industry, private sector and government).
- ✓ Identifying emerging job roles and the skill sets that will be most in demand in the near future.
- ✓ Translating findings into actionable recommendations for learners, trainers, employers and policymakers.

◆ 4. Scope of the Study

The study is deliberately structured around a $5 \times 5 \times 3$ framework, combining five sectors, five lenses and three stakeholder personas to ensure the findings are both broad in coverage and sharp in focus.

Manufacturing · Healthcare · Infrastructure · Energy · FMCG	Artificial Intelligence · DPI · Climate & Green Jobs · Workforce Mobility · Industry 4.0	Employer (industry) · Learner (education / up-skilling / re-skilling) · Trainer (industry, private sector, government)

Geographically, the study is anchored in India but benchmarks each sector against leading global practices (e.g., Germany and Japan in manufacturing, Taiwan and South Korea in healthcare, the UK and Australia in infrastructure planning, Denmark and Germany in green energy). Time-wise, the primary lens is a 2–3 year forward outlook for job creation, reskilling and upskilling.

🧠 5. Key Features of the Report

- ✓ **Five sector deep-dives** with macroeconomic data, employment scenarios and government schemes.
- ✓ **Primary survey insights** from over 200 industry leaders, learners, trainers and policymakers.
- ✓ **International benchmarking** against comparable economies and best practices for each sector.
- ✓ **Emerging job-role catalogues** and in-demand skill sets, sector by sector.
- ✓ **A dedicated Digital Public Infrastructure (DPI) playbook** — covering centralized registries, digital verifiable credentials, skilling passports, portable skills, social protection and AI knowledge assistants.
- ✓ **A Harvey-ball impact matrix** scoring AI, DPI, Climate, Workforce Mobility and Industry 4.0 against every sector and stakeholder persona.
- ✓ **Actionable recommendations** for learners, trainers, employers and government policymakers.

6. Technologies & Themes Analysed

Rather than a single technology stack, FOJ 3.0 analyses a basket of converging technological and policy themes that are reshaping the world of work:

Artificial Intelligence & Generative AI	Predictive analytics, automation, AI-enabled diagnostics, supply-chain optimisation, and the reskilling needs they create.
Digital Public Infrastructure (DPI) / Digital Public Goods (DPG)	Population-scale digital rails such as Aadhaar, UPI and the Skill India Digital Hub (SIDH) that power identity, credentialing and service delivery.
Industry 4.0	IoT, robotics, big data & analytics, cloud computing, additive manufacturing (3D printing) and cybersecurity.
Green Technology & Climate Action	Renewable energy, sustainable manufacturing, green building, circular economy and ESG-aligned operations.
Workforce Mobility Platforms	International mobility corridors, skills-equivalency mapping and cross-cultural competency for a globally deployable workforce.

7. Who This Report Is For

-  **Policymakers** shaping skilling, education and employment reform at the central and state level.
-  **Business leaders and HR/CHRO functions** planning workforce strategy across the five focus sectors.
-  **Learners** — students, early-career professionals and workers seeking reskilling or upskilling.
-  **Trainers, academic institutions and ITIs** designing curricula aligned with industry needs.
-  **Skilling ecosystem partners**, including NSDC, Sector Skill Councils and CSR/skilling foundations.

8. Expected Outcomes

The report is designed to serve as a crucial guide for policymakers, business leaders and individuals alike. Its intended outcomes include:

-  A shared, evidence-based understanding of which job roles and skills will matter most over the next 2–3 years across Manufacturing, Healthcare, Infrastructure, Energy and FMCG.
-  A practical framework (the 5 lenses × 5 sectors × 3 personas model) that learners, trainers and employers can use to benchmark their own readiness.
-  Clear policy recommendations — from revamping ITI infrastructure to strengthening industry–academia linkages and scaling Digital Public Infrastructure for skilling.
-  A foundation for reimagining career paths, encouraging lifelong learning, and building an inclusive, climate-ready and globally mobile workforce.

Continue to Part 2 — Infographics & Diagrams for the visual data behind these findings, and Part 3 — Brief Summary for a condensed executive overview.